

DPU ACS College Pimpri Pune

Advanced Genomics

Practical No. – 3

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Roll no. 03

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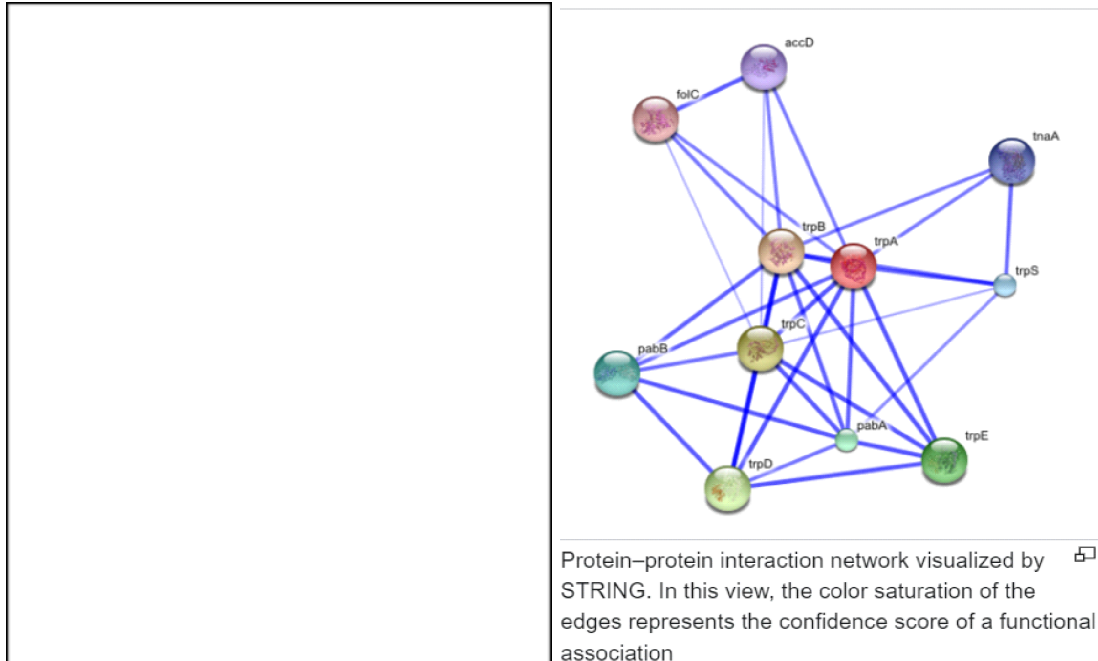
Q.1 Explore the STRING database and write report on it.

Ans : In molecular biology, STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) is a biological database and web resource of known and predicted protein–protein interactions. The STRING database contains information from numerous sources, including experimental data, computational prediction methods and public text collections. It is freely accessible and it is regularly updated. The resource also serves to highlight functional enrichments in user-provided lists of proteins, using a number of functional classification systems such as GO, Pfam and KEGG. The latest version 11b contains information on about 24,5 million proteins from more than 5000 organisms.

Protein–protein interaction networks are an important ingredient for the system-level understanding of cellular processes. Such networks can be used for filtering and assessing functional genomics data and for providing an intuitive platform for annotating structural, functional and evolutionary properties of proteins.

Features:

The data is weighted and integrated and a confidence score is calculated for all protein interactions. Results of the various computational predictions can be inspected from different designated views. There are two modes of STRING: Protein-mode and COG-mode. Predicted interactions are propagated to proteins in other organisms for which interaction has been described by inference of orthology. A web interface is available to access the data and to give a fast overview of the proteins and their interactions.



Data sources:

Like many other databases that store protein association knowledge, STRING imports data from experimentally derived protein-protein interactions through literature curation. Furthermore, STRING also store computationally predicted interactions from: (i) text mining of scientific texts, (ii) interactions computed from genomic features, and (iii) interactions transferred from model organisms based on orthology.

Q.2 Search the following proteins and

1) Amyloid :-

1. Number of nodes: 21
2. Number of edges: 88
3. Average node degree: 8.38
4. Avg. local clustering coefficient: 0.791
5. Expected number of edges: 24
6. PPI enrichment p-value: $< 1.0e-16$

2) Tau :-

1. Number of nodes: 11
2. Number of edges: 23
3. Average node degree: 4.18
4. Avg. local clustering coefficient: 0.763
5. Expected number of edges: 11
6. PPI enrichment p-value: 0.00155

Q.3 Enlist the Functional Enrichments in the Networks.

1) Amyloid :



> Biological Process (Gene Ontology)				
<i>GO-term</i>	<i>description</i>	<i>count in network</i>	<i>strength</i>	<i>false discovery rate</i>
GO:0051563	Smooth endoplasmic reticulum calcium ion homeostasis	2 of 2	2.97	0.0017
GO:0034205	Amyloid-beta formation	7 of 9	2.86	4.51e-14
GO:1905908	Positive regulation of amyloid fibril formation	3 of 4	2.85	3.00e-05
GO:0007220	Notch receptor processing	6 of 9	2.8	1.04e-11
GO:1902996	Regulation of neurofibrillary tangle assembly	2 of 3	2.8	0.0026
(more ...)				
> Molecular Function (Gene Ontology)				
<i>GO-term</i>	<i>description</i>	<i>count in network</i>	<i>strength</i>	<i>false discovery rate</i>
GO:0042500	Aspartic endopeptidase activity, intramembrane cleaving	3 of 8	2.55	0.00014
GO:1904399	Heparan sulfate binding	2 of 6	2.5	0.0135
GO:0061133	Endopeptidase activator activity	3 of 13	2.34	0.00036
GO:0004190	Aspartic-type endopeptidase activity	4 of 27	2.14	0.00013
GO:0050750	Low-density lipoprotein particle receptor binding	3 of 25	2.05	0.0018
(more ...)				
> Cellular Component (Gene Ontology)				
<i>GO-term</i>	<i>description</i>	<i>count in network</i>	<i>strength</i>	<i>false discovery rate</i>
GO:0070765	Gamma-secretase complex	7 of 7	2.97	3.54e-15
GO:0034363	Intermediate-density lipoprotein particle	2 of 5	2.57	0.00098
GO:0035253	Ciliary rootlet	3 of 13	2.34	3.94e-05
GO:0005641	Nuclear envelope lumen	2 of 10	2.27	0.0025
GO:0042589	Zymogen granule membrane	2 of 11	2.23	0.0030
(more ...)				



Reference publications (PubMed)				
publication	(year) title	count in network	strength	false discovery rate
PMID:22122073	(2012) Assembly of the presenilin gamma-epsilon-secretas...	8 of 8	2.97	7.40e-16
PMID:30944650	(2019) Evaluation of the prognostic significances of gamma...	7 of 7	2.97	8.18e-14
PMID:22654105	(2012) Nuclear factor-kappaB regulates betaAPP and beta- ...	7 of 7	2.97	8.18e-14
PMID:19159235	(2009) Purification, pharmacological modulation, and bioch...	7 of 7	2.97	8.18e-14
PMID:15766275	(2005) Characterization of the reconstituted gamma-secret...	7 of 7	2.97	8.18e-14
(more ...)				
Local network cluster (STRING)				
cluster	description	count in network	strength	false discovery rate
CL:21512	Gamma-secretase complex	6 of 6	2.97	5.60e-13
CL:21509	Amyloid-beta formation, and Memapsin-like	8 of 11	2.83	3.09e-17
CL:13364	Transmembrane emp24 domain-containing protein	3 of 6	2.67	6.64e-05
CL:21507	Mixed, incl. Amyloid-beta metabolic process, and Amyloid A4	10 of 24	2.59	8.55e-20
CL:21538	Amyloid A4, and Amyloid beta precursor protein binding fa...	2 of 5	2.57	0.0115
(more ...)				
KEGG Pathways				
pathway	description	count in network	strength	false discovery rate
hsa04330	Notch signaling pathway	6 of 52	2.03	4.37e-09
hsa05010	Alzheimer disease	10 of 354	1.42	4.02e-10
Reactome Pathways				
pathway	description	count in network	strength	false discovery rate
HSA-5619070	Defective SLC16A1 causes symptomatic deficiency in lactat...	2 of 2	2.97	0.00048
HSA-9017802	Noncanonical activation of NOTCH3	6 of 8	2.85	1.91e-12
HSA-9013700	NOTCH4 Activation and Transmission of Signal to the Nucle...	6 of 11	2.71	3.57e-12
HSA-193692	Regulated proteolysis of p75NTR	6 of 11	2.71	3.57e-12
HSA-205043	NRIF signals cell death from the nucleus	6 of 16	2.55	1.12e-11
(more ...)				

2) Tau

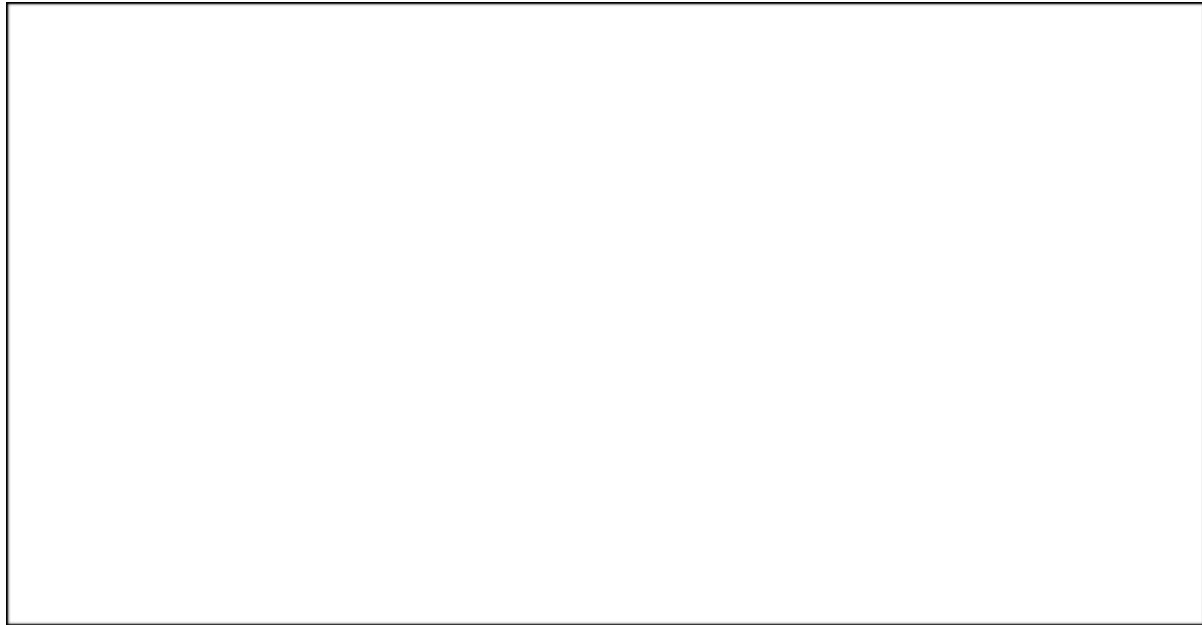


Functional enrichments in your network

Note: some enrichments may be expected here ([why?](#))

[explain columns](#)

> Biological Process (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0043524	Negative regulation of neuron apoptotic process	2 of 19	2.13	0.0234
GO:0048489	Synaptic vesicle transport	2 of 22	2.06	0.0266
GO:1901215	Negative regulation of neuron death	3 of 34	2.05	0.0021
GO:0046627	Negative regulation of insulin receptor signaling pathway	2 of 27	1.97	0.0303
GO:0070507	Regulation of microtubule cytoskeleton organization	3 of 47	1.91	0.0035
(more ...)				
Molecular Function (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0140096	Catalytic activity, acting on a protein	7 of 1294	0.84	0.0333
Cellular Component (Gene Ontology)				
GO-term	description	count in network	strength	false discovery rate
GO:0031594	Neuromuscular junction	3 of 85	1.65	0.0158
GO:0030424	Axon	5 of 250	1.4	0.0010
> Reference publications (PubMed)				
publication	(year) title	count in network	strength	false discovery rate
PMID:29621183	(2018) Tau-Induced Pathology in Epilepsy and Dementia: No...	3 of 3	3.1	2.23e-05
PMID:36632700	(2023) The role of STING signaling in central nervous syste...	2 of 2	3.1	0.0013
PMID:36293118	(2022) Radiation as a Tool against Neurodegeneration-A Po...	2 of 2	3.1	0.0013
PMID:35335141	(2022) Amyloid Cross-Seeding: Mechanism, Implication, an...	2 of 2	3.1	0.0013
PMID:33642981	(2021) DI-3-n-Butylphthalide Reduces Cognitive Deficits and ...	2 of 2	3.1	0.0013
(more ...)				
Local network cluster (STRING)				
cluster	description	count in network	strength	false discovery rate
CL:8275	Mixed, incl. Stathmin family, and Intermediate filament-base...	3 of 6	2.8	4.63e-05
CL:8259	Mixed, incl. Axonal transport of mitochondrion, and Kinesin ...	4 of 14	2.56	1.93e-06



Reactome Pathways				
pathway	description	count in network	strength	false discovery rate
DME-264870	Caspase-mediated cleavage of cytoskeletal proteins	2 of 7	2.56	0.0283
DME-399956	CRMPs in Sema3A signaling	2 of 13	2.29	0.0412

> Subcellular localization (COMPARTMENTS)				
compartment	description	count in network	strength	false discovery rate
GOCC:0106003	Amyloid-beta complex	4 of 4	3.1	1.13e-08
GOCC:0097418	Neurofibrillary tangle	3 of 3	3.1	3.74e-06
GOCC:0005883	Neurofilament	2 of 2	3.1	0.00075
GOCC:0097413	Lewy body	2 of 5	2.71	0.0015
GOCC:0016234	Inclusion body	5 of 17	2.57	4.23e-09
(more ...)				

> Drosophila Phenotype (Monarch)				
phenotype	description	count in network	strength	false discovery rate
FBcv:0002030	Lethal - all die during pupal stage	2 of 90	1.45	0.0478
FBcv:0000449	Abnormal planar polarity	3 of 202	1.27	0.0198
FBcv:0000385	Short lived	9 of 1170	0.99	1.68e-06
FBcv:0000425	Increased cell death	4 of 554	0.96	0.0224
FBcv:0000434	Abnormal neurophysiology	4 of 638	0.9	0.0271
(more ...)				

Q.4 Enlist the Node2 accession numbers with total count.

1) Amyloid

node1	node2	node1 accession	node2 accession	node1 annotation	node2 annotation	score
APH1A	APBB1	ENSP00000358105	ENSP00000477213	Gamma-secretase subunit APH-1A; Non-...	Amyloid-beta A4 precursor protein-bindin...	0.442
APOE	APBB1	ENSP00000252486	ENSP00000477213	Apolipoprotein E; APOE is an apolipoprot...	Amyloid-beta A4 precursor protein-bindin...	0.623
APP	APBB1	ENSP00000284981	ENSP00000477213	Gamma-secretase C-terminal fragment 5...	Amyloid-beta A4 precursor protein-bindin...	0.999
BACE1	APBB1	ENSP00000318585	ENSP00000477213	Beta-secretase 1; Responsible for the pro...	Amyloid-beta A4 precursor protein-bindin...	0.791
GSAP	APBB1	ENSP00000257626	ENSP00000477213	Gamma-secretase-activating protein 16 k...	Amyloid-beta A4 precursor protein-bindin...	0.533
KAT5	APBB1	ENSP00000340330	ENSP00000477213	Histone acetyltransferase KAT5; Catalytic...	Amyloid-beta A4 precursor protein-bindin...	0.999
NCSTN	APBB1	ENSP00000294785	ENSP00000477213	Nicastrin; Essential subunit of the gamm...	Amyloid-beta A4 precursor protein-bindin...	0.535
PSEN1	APBB1	ENSP00000326366	ENSP00000477213	Presenilin-1 CTF subunit; Catalytic subuni...	Amyloid-beta A4 precursor protein-bindin...	0.703
PSEN2	APBB1	ENSP00000355747	ENSP00000477213	Presenilin-2 CTF subunit; Probable catalyt...	Amyloid-beta A4 precursor protein-bindin...	0.588
PSENEN	APBB1	ENSP00000468411	ENSP00000477213	Gamma-secretase subunit PEN-2; Essenti...	Amyloid-beta A4 precursor protein-bindin...	0.456
SORL1	APBB1	ENSP00000260197	ENSP00000477213	Sortilin-related receptor; Sorting receptor ...	Amyloid-beta A4 precursor protein-bindin...	0.478
APBB1	APH1A	ENSP00000477213	ENSP00000358105	Amyloid-beta A4 precursor protein-bindin...	Gamma-secretase subunit APH-1A; Non-...	0.442
APH1B	APH1A	ENSP00000261879	ENSP00000358105	Gamma-secretase subunit APH-1B; Prob...	Gamma-secretase subunit APH-1A; Non-...	0.975
APOE	APH1A	ENSP00000252486	ENSP00000358105	Apolipoprotein E; APOE is an apolipoprot...	Gamma-secretase subunit APH-1A; Non-...	0.431
APP	APH1A	ENSP00000284981	ENSP00000358105	Gamma-secretase C-terminal fragment 5...	Gamma-secretase subunit APH-1A; Non-...	0.986
BACE1	APH1A	ENSP00000318585	ENSP00000358105	Beta-secretase 1; Responsible for the pro...	Gamma-secretase subunit APH-1A; Non-...	0.793
BSG	APH1A	ENSP00000333769	ENSP00000358105	Basigin; Plays an important role in targeti...	Gamma-secretase subunit APH-1A; Non-...	0.564
GSAP	APH1A	ENSP00000257626	ENSP00000358105	Gamma-secretase-activating protein 16 k...	Gamma-secretase subunit APH-1A; Non-...	0.757
NCSTN	APH1A	ENSP00000294785	ENSP00000358105	Nicastrin; Essential subunit of the gamm...	Gamma-secretase subunit APH-1A; Non-...	0.999
PSEN1	APH1A	ENSP00000326366	ENSP00000358105	Presenilin-1 CTF subunit; Catalytic subuni...	Gamma-secretase subunit APH-1A; Non-...	0.999

Total count of Node2: 176

<u>node2 accession</u>		<u>node2 accession</u>		<u>node2 accession</u>
ENSP00000294785		ENSP00000326366		ENSP00000468411
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ENSP00000419425		ENSP00000355747		ENSP00000468411
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ENSP00000419425		ENSP00000355747		ENSP00000260197
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ENSP00000262225				
ENSP00000262225				
ENSP00000330945				
ENSP00000330945				
ENSP00000353785				
ENSP00000353785				

2) Tau

<u>node1</u>	<u>node2</u>	<u>node1 accession</u>	<u>node2 accession</u>	<u>node1 annotation</u>	<u>node2 annotation</u>	<u>score</u>
Axn	Apc2	FBpp0084919	FBpp0401579	Axin (Axn) encodes the key scaffolding p...	Adenomatous polyposis coli 2 (Apc2) en...	0.999
Dredd	Apc2	FBpp0113052	FBpp0401579	Caspase-8 subunit p10; Effector of the pr...	Adenomatous polyposis coli 2 (Apc2) en...	0.750
arm	Apc2	FBpp0089032	FBpp0401579	Armadillo segment polarity protein; Isofo...	Adenomatous polyposis coli 2 (Apc2) en...	0.998
dsh	Apc2	FBpp0073310	FBpp0401579	Segment polarity protein dishevelled; Req...	Adenomatous polyposis coli 2 (Apc2) en...	0.957
sgg	Apc2	FBpp0304140	FBpp0401579	Protein kinase shaggy; Required for sever...	Adenomatous polyposis coli 2 (Apc2) en...	0.988
slmb	Apc2	FBpp0303082	FBpp0401579	Supernumerary limbs (slmb) encodes an ...	Adenomatous polyposis coli 2 (Apc2) en...	0.724
Apc2	Axn	FBpp0401579	FBpp0084919	Adenomatous polyposis coli 2 (Apc2) en...	Axin (Axn) encodes the key scaffolding p...	0.999
N	Axn	FBpp0070483	FBpp0084919	Processed neurogenic locus Notch protel...	Axin (Axn) encodes the key scaffolding p...	0.564
arm	Axn	FBpp0089032	FBpp0084919	Armadillo segment polarity protein; Isofo...	Axin (Axn) encodes the key scaffolding p...	0.997
dsh	Axn	FBpp0073310	FBpp0084919	Segment polarity protein dishevelled; Req...	Axin (Axn) encodes the key scaffolding p...	0.999
sgg	Axn	FBpp0304140	FBpp0084919	Protein kinase shaggy; Required for sever...	Axin (Axn) encodes the key scaffolding p...	0.999
slmb	Axn	FBpp0303082	FBpp0084919	Supernumerary limbs (slmb) encodes an ...	Axin (Axn) encodes the key scaffolding p...	0.999
TBPH	CG14074	FBpp0088568	FBpp0300969	TAR DNA-binding protein-43 homolog (T...	IP10578p.	0.994
lwr	CG14074	FBpp0301731	FBpp0300969	Lesswright (lwr) encodes Ubc9, a SUMO ...	IP10578p.	0.994
Gsc	CG32846	FBpp0309910	FBpp0293005	Goosecoid (Gsc) encodes a homeoprotei...	Uncharacterized protein, isoform B.	0.994
TBPH	CG32846	FBpp0088568	FBpp0293005	TAR DNA-binding protein-43 homolog (T...	Uncharacterized protein, isoform B.	0.994
Cdk5alpha	Cdk5	FBpp0304060	FBpp0086483	Cdk5 activator-like protein (Cdk5alpha) e...	Cyclin-dependent kinase 5 (Cdk5) encode...	0.998
Nct	Cdk5	FBpp0306637	FBpp0086483	Nicastrin; Essential subunit of the gamm...	Cyclin-dependent kinase 5 (Cdk5) encode...	0.548
Psn	Cdk5	FBpp0077850	FBpp0086483	Presenilin homolog; Probable catalytic su...	Cyclin-dependent kinase 5 (Cdk5) encode...	0.842
STUB1	Cdk5	FBpp0079617	FBpp0086483	Carboxy terminus of Hsp70-interacting pr...	Cyclin-dependent kinase 5 (Cdk5) encode...	0.417

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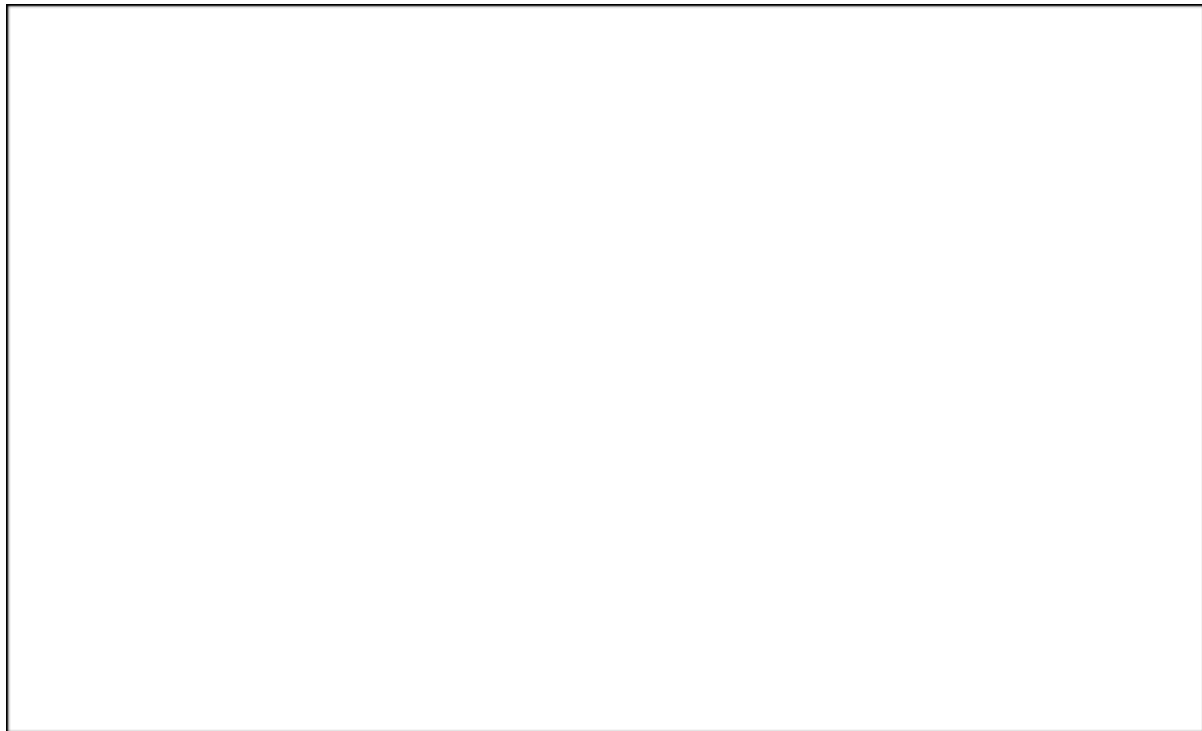
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	FBpp0079462			

Count 176

Q.5 K-means clustering with 5 numbers of clusters and no edge between them.

1) Amyloid:



Clustering Options

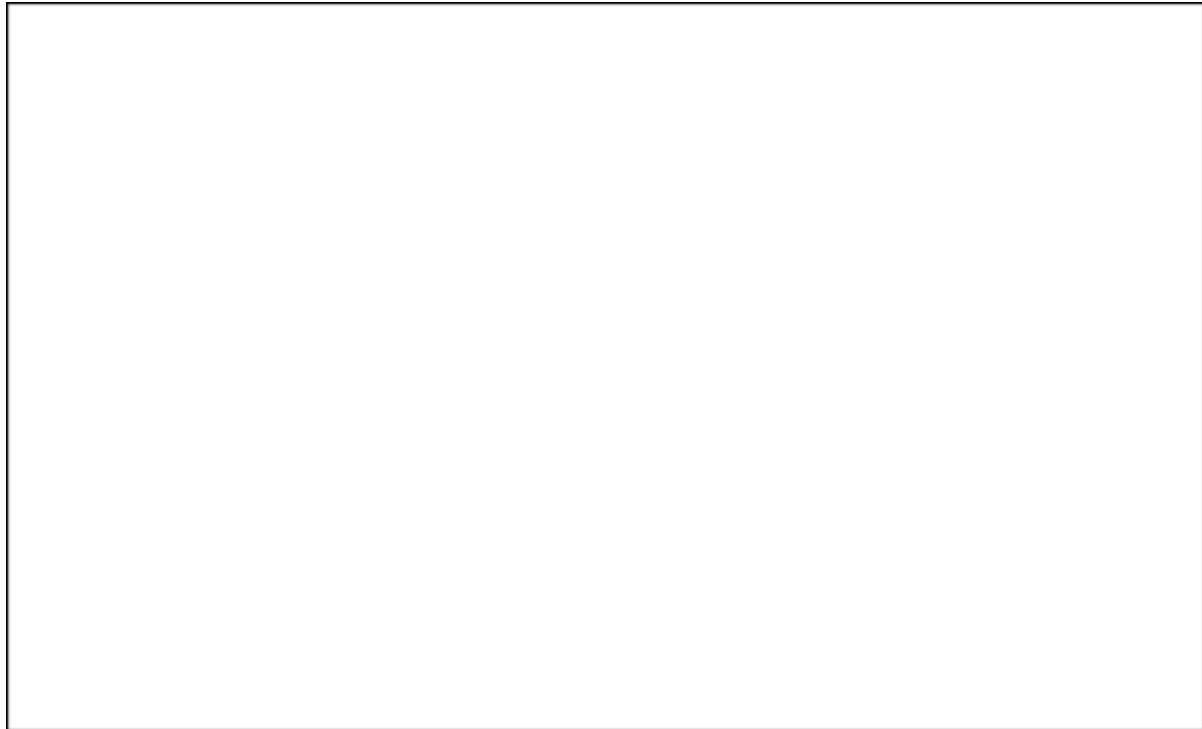
- ☐ no clustering (network is shown as it is)
- ☒ kmeans clustering (network is clustered to a specified number of clusters)
- number of clusters:
- edges between clusters:
- ☐ MCL clustering (network is clustered to a specified "MCL inflation parameter")

APPLY

Clusters

bubble	cluster Id	gene count	protein names
	Cluster 1	11	APBB1,APH1B,APOE,APP,BACE1,GSAP,KAT5,NCSTN,PSEN1,PSEN2,SORL1
	Cluster 2	6	APH1A,PSENEN,TMED10,TMED2,TMED9,TPPP
	Cluster 3	2	BSG,PPIA
	Cluster 4	1	NEURL1B
	Cluster 5	1	SLC16A1

2) Tau:



Clustering Options

- ☐ no clustering (network is shown as it is)
- ☒ kmeans clustering (network is clustered to a specified number of clusters)
- number of clusters:
- edges between clusters:
- ☐ MCL clustering (network is clustered to a specified "MCL inflation parameter")

APPLY

Clusters

bubble	cluster Id	gene count	protein names
	Cluster 1	5	Cdk5,STUB1,otk,tau,und
	Cluster 2	5	N,Nct,Psn,aph-1,pen-2
	Cluster 3	9	CG14074,CG32846,Gsc,Sod1,TBPH,apt,futsch,htt,lwr
	Cluster 4	8	Apc2,Axn,Cdk5alpha,arm,dsh,sgg,slmb,tim
	Cluster 5	4	Diap2,Dredd,Fadd,imd

Q.6 Explore the following features of STRING database and interpret them with images of proteins.

Ans: - The STRING database is a comprehensive resource that provides a global perspective on protein-protein interactions. Here are some of its key features:

1. **Protein-Protein Interaction Networks:** STRING database provides functional protein association networks. These networks represent predicted associations between proteins.
2. **Functional Enrichment Analysis:** This feature allows users to identify statistically overrepresented functional categories in a set of proteins.
3. **Multiple Search Options:** Users can search STRING by single protein name, multiple names, or by amino acid sequence. There's also the possibility to search by protein family.
4. **Network View:** The network view summarizes the network of predicted associations for a particular group of proteins. The nodes are proteins and the edges represent the predicted functional associations.
5. **Evidence Types:** In evidence mode, an edge may be drawn with up to 7 differently colored lines, each representing a type of evidence used in predicting the associations.
6. **Confidence Score:** The data is weighted and integrated, and a confidence score is calculated for all protein interactions.
7. **Customization and Usability Features:** STRING focuses on usability features such as customization, enrichment detection, and programmatic access.

As for images of proteins, they are typically represented as complex structures, often visualized in 3D models. These images are usually created using data from techniques like X-ray crystallography or nuclear magnetic resonance (NMR) spectroscopy. They can show the intricate details of protein structures, including the arrangement of individual atoms, the folding of the protein chain, and the location of active sites.

tau

Information

Microtubule-associated protein; Microtubule binding. It is involved in the biological process described with: neuron projection development; rhabdome morphogenesis; negative regulation of neuron death; microtubule cytoskeleton organization.

Identifier: FBpp0311078, tau

Organism: *Drosophila melanogaster*

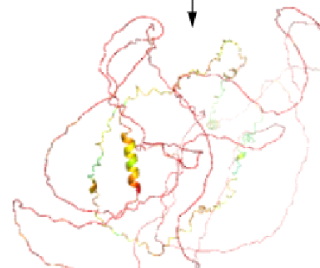
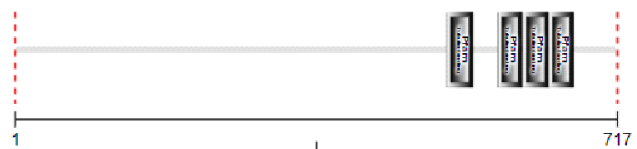
[e!](#) [e!](#) [FlyBase](#) [Kegg](#) [RefSeq](#)



- [show protein sequence](#)
- [homologs among STRING organisms](#)

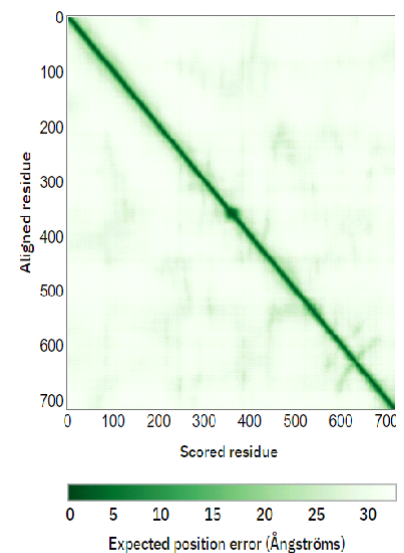
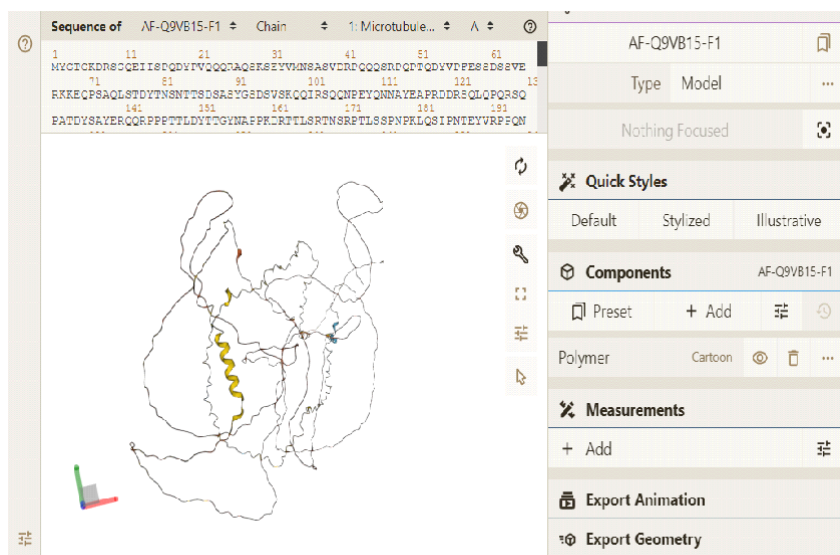
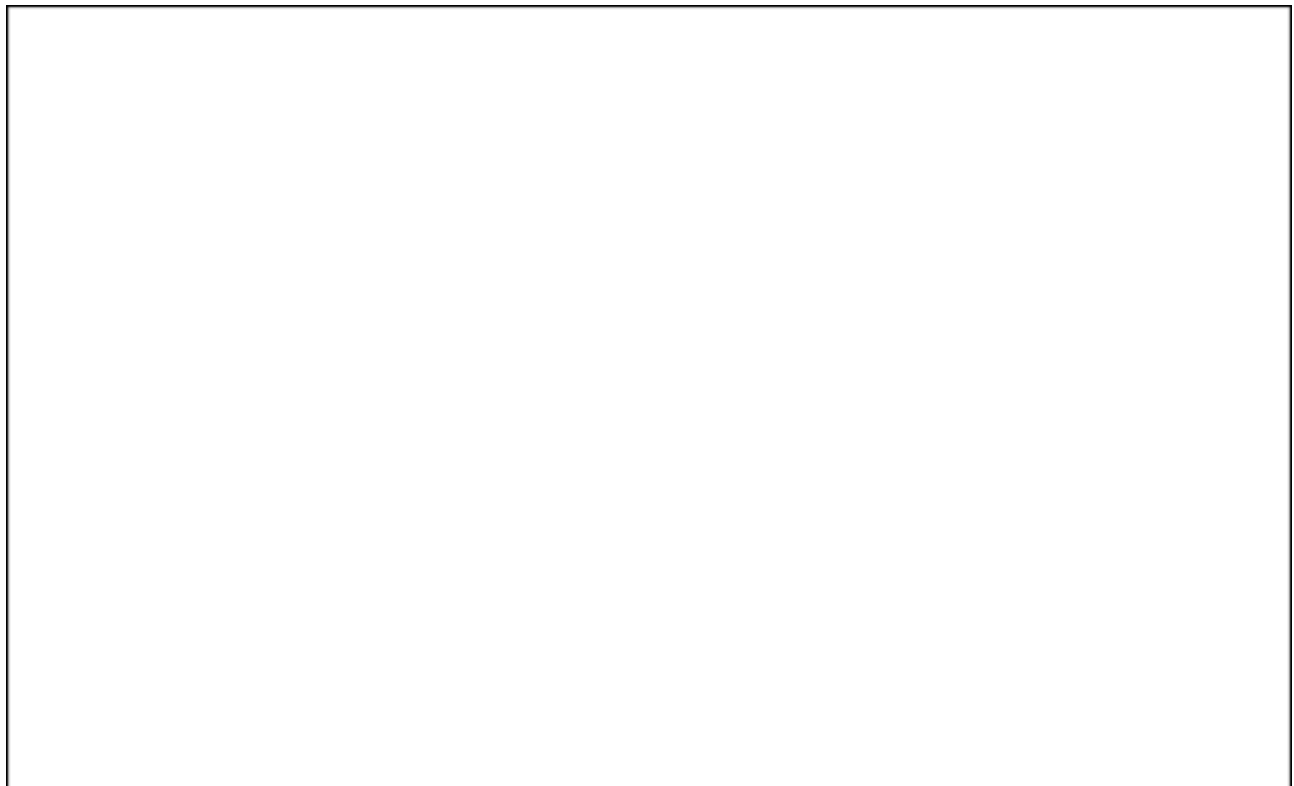
[Enable node coloring mode](#)

[Show this node's terms in the analysis table](#)



AlphaFold model (Q9VB15)
identity: 100%

SS



Model Confidence

- Very high (pLDDT > 90)
- High (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

AlphaFold produces a per-residue model confidence score (pLDDT) between 0 and 100. Some regions below 50 pLDDT may be unstructured in isolation.

Click and drag a box on the PAE viewer to select regions of the structure and highlight them on the 3D viewer.

PAE data is useful for assessing inter-domain accuracy – [go to Help section below](#) for more information.

